AAE 204 – Aeromechanics II

3 Credits

1. Course Information/Description:

This course introduces aerospace engineering students to the mechanics of solids, concepts of force/stress/equilibrium, deformation/strain/compatibility, and stress/strain material behaviors. These concepts, through lecture material, classroom examples and course work (homework assignments and design projects), are applied to elementary aerospace structural components of rods in tension and compression, shafts in torsion, beams in bending and shear, and thin-walled vessels under pressure.

2. Brightspace Learning Management System:

Course videos, materials, and resources are available through Purdue's Brightspace system; students should begin with the Start Here tab to familiarize themselves with course organization, navigation, and available campus support resources.

3. Textbook:

The primary textbook for this course is *Mechanics of Materials*, 8th Edition, by J. M. Gere and B. J. Goodno (ISBN: 1-111-57773-0). A supplementary reference is *Mechanics of Materials*, 8th Edition, by Ferdinand P. Beer, E. Russell Johnston, John T. DeWolf, and David Mazurek (ISBN: 978-1-260-11327-3, McGraw-Hill).

4. Attendance Policy:

Students are expected to attend all scheduled classes in accordance with Purdue University policy. Attendance contributes 2% of the final grade and is measured through 10 unannounced iClicker checks during the semester, with missed checks proportionally reducing the attendance score.

5. Assignments:

To support student learning and evaluate understanding throughout the course, a variety of assignments and assessments will be administered. These are designed to provide ongoing practice, reinforce key concepts, and measure both conceptual knowledge and problem-solving skills. The breakdown of assignments and their contribution to the final grade is as follows:

Assignments and Exams	Due	Points (Traditional Section)	Points (Honors Section)
Homework	Weekly	20	20
Attendance	10 counts	2	2
Exam 1	Monday 02/23	25	22
Exam 2	Monday 03/30	25	22
Exam 3	TBD	28	24
Project		0	10
		Total: 100	Total: 100

6. Extra Credit:

Extra credit of 1% will be given to students who complete both mid-term and final course evaluations.

7. Course Learning Objectives:

On completing this course, you shall be able to:

- Describe the material behavior of deformable bodies when subjected to loads.
- Create and use free-body diagrams to compute resultant support forces and moments.
- Compute internal force and moment distributions.
- Properly select and use materials based upon their mechanical properties.
- Compute stresses, strains, deformation, and displacements of basic structural components.
- Predict structural integrity based on calculations.
- Apply these ideas for analysis of components related to aerospace applications.

8. Evaluation:

Your grade is based on your performance in the homework and exams. Assigned letter grade cutoffs are as follows:

Percentage	Letter grade	Percentage	Letter grade
96.67 – 100	A+	73.33 – 76.66	C
93.33 – 96.66	A	70 – 73.32	C-
90 – 93.32	A-	66.67 – 69.99	D+
86.67 – 89.99	B+	63.33 – 66.66	D
83.33 – 86.66	B	60 – 63.32	D-
80 – 83.32	B-	0 – 59.99	F
76.67 – 79.99	C+		

Please note that final grades are conclusive and will not be changed under any circumstances.

1 Evaluation:

AI in this course is allowed for purposes like finding/creating new exercises applicable to the course materials. AI tools can’t be used for partially or entirely solving homework assignments